

Longitudinal Propensity Score Analysis: Method Comparison

J Kyle Armstrong, PhD

2026-07-05

Table of contents

1	Objective	1
1.1	Outcome Type Interpretations	1
2	Representative Scenario (Rho=0.5, Adherence ~75%)	3
2.1	Patient Level Comparisons	3
2.2	Observation Level Comparisons	3
3	Method Comparison (Pooled)	5
4	Visualizing Bias and Coverage	7
4.1	Bias vs Confounding Strength (Rho Sweep)	7
4.2	Bias vs Observed Adherence Rate (Adherence Sweep)	7
5	Longitudinal Estimates Distribution	10
6	Discussion	11

1 Objective

This report compares treatment effect estimates across various analytic strategies and scenarios of confounding (ρ_{uv}). We explore variations of the populations (Intention-to-Treat (ITT), Per-Protocol (PP), and Propensity Score Matching (PSM)) across both visit-level and patient-level adherence.

1.1 Outcome Type Interpretations

The simulation framework dynamically adapts to the selected outcome type (continuous). Depending on the configuration, the results in this report should be interpreted as follows:

- **Continuous Outcomes:** Treatment effects are modeled using standard linear/mixed-effects models. The **Estimate** (and Bias) represents the absolute difference in expected means between the treatments.
- **Binary Outcomes:** Treatment effects are modeled using logistic regression with binomial link functions. The **Estimate** (and Bias) is interpreted on the log-odds scale, and the estimates represent log-odds ratios.

2 Representative Scenario ($\rho=0.5$, Adherence $\sim 75\%$)

2.1 Patient Level Comparisons

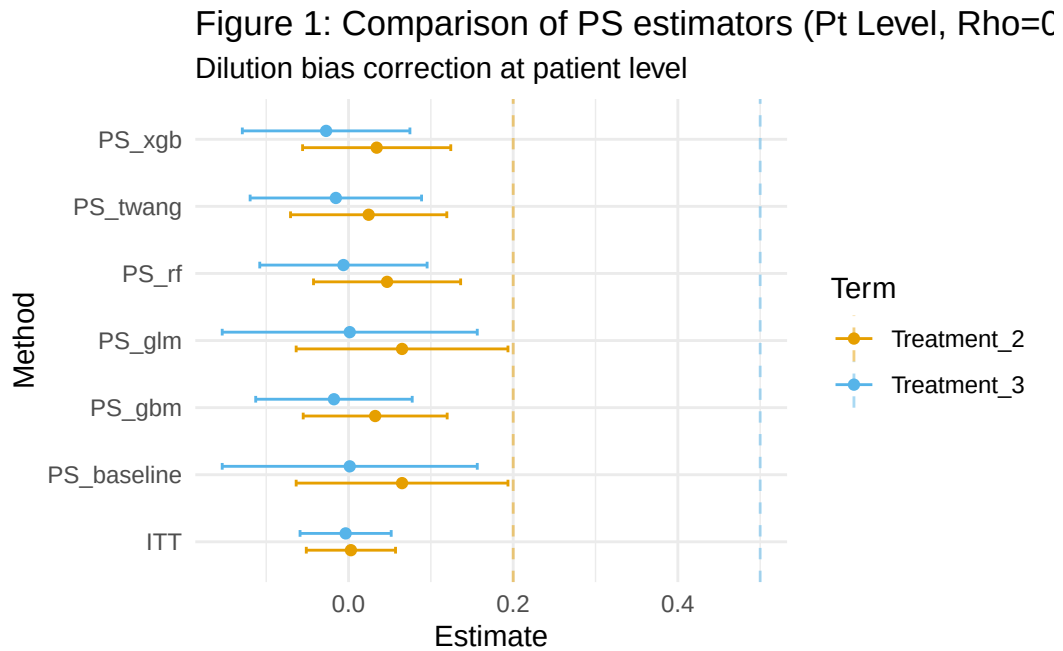


Figure 1

2.2 Observation Level Comparisons

Figure 2: ITT vs Naive vs PS (Pt Level, $\text{Rho}=0.5$)

Selection bias adjustment at patient level

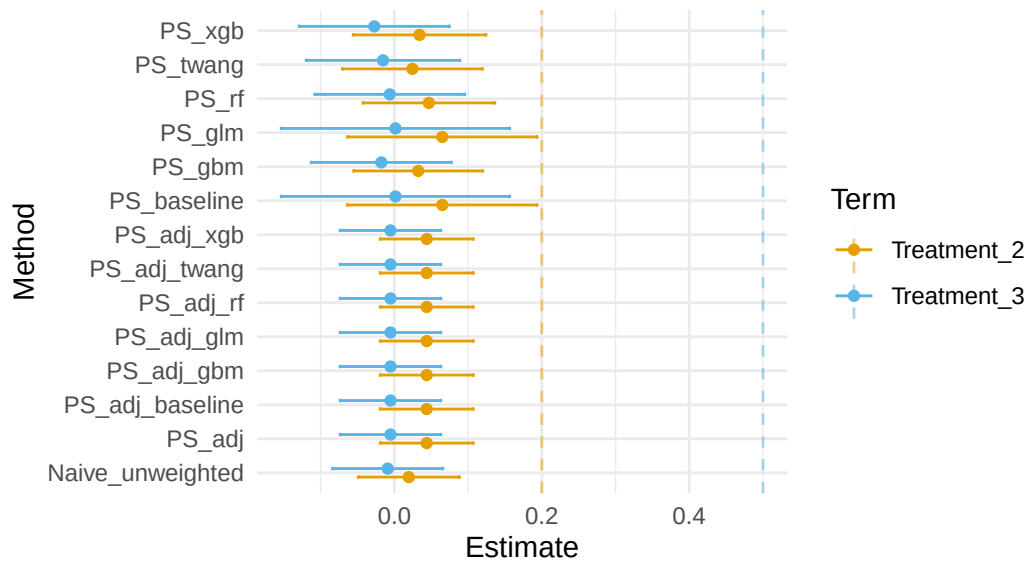


Figure 2

Figure 3: PP(obs) vs PS(obs) (Obs Level, $\text{Rho}=0.5$)

Selection bias adjustment at observation level

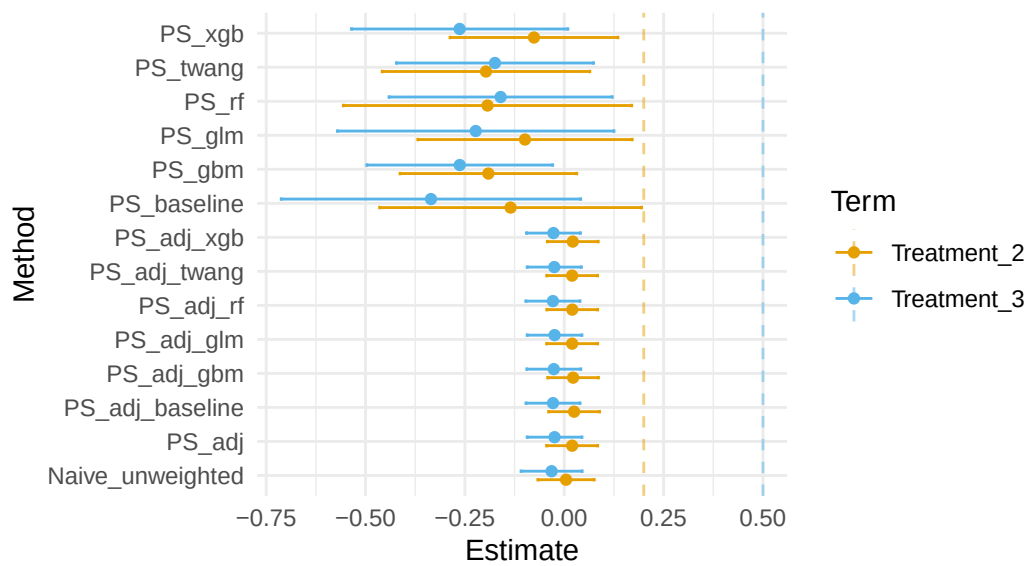


Figure 3

3 Method Comparison (Pooled)

Below we compare the estimated treatment effects across all scenarios.

Table 1: Averaged Treatment Effects Across Scenarios (including Twang)

Level	Method	Term	Mean_Estimate	SD_Estimate
obs	ITT	Treatment_2	-0.002	0.021
obs	Naive_unweighted	Treatment_2	-0.023	0.300
obs	PS_adj	Treatment_2	-1.071	7.333
obs	PS_adj_baseline	Treatment_2	-1.083	7.331
obs	PS_adj_gbm	Treatment_2	-1.084	7.330
obs	PS_adj_glm	Treatment_2	-1.071	7.333
obs	PS_adj_rf	Treatment_2	-1.084	7.331
obs	PS_adj_twang	Treatment_2	-1.071	7.333
obs	PS_adj_xgb	Treatment_2	-1.083	7.331
obs	PS_baseline	Treatment_2	0.014	0.432
obs	PS_gbm	Treatment_2	0.009	0.382
obs	PS_glm	Treatment_2	-0.021	0.412
obs	PS_rf	Treatment_2	-0.035	0.405
obs	PS_twang	Treatment_2	-0.025	0.394
obs	PS_xgb	Treatment_2	-0.022	0.395
pt	ITT	Treatment_2	NA	NA
pt	Naive_unweighted	Treatment_2	NA	NA
pt	PS_adj	Treatment_2	NA	NA
pt	PS_adj_baseline	Treatment_2	NA	NA
pt	PS_adj_gbm	Treatment_2	NA	NA
pt	PS_adj_glm	Treatment_2	NA	NA
pt	PS_adj_rf	Treatment_2	NA	NA
pt	PS_adj_twang	Treatment_2	NA	NA
pt	PS_adj_xgb	Treatment_2	NA	NA
pt	PS_baseline	Treatment_2	NA	NA
pt	PS_gbm	Treatment_2	NA	NA
pt	PS_glm	Treatment_2	NA	NA
pt	PS_rf	Treatment_2	NA	NA
pt	PS_twang	Treatment_2	NA	NA
pt	PS_xgb	Treatment_2	NA	NA
obs	ITT	Treatment_3	0.000	0.026
obs	Naive_unweighted	Treatment_3	-0.007	0.179
obs	PS_adj	Treatment_3	-0.331	1.372
obs	PS_adj_baseline	Treatment_3	-0.323	1.374
obs	PS_adj_gbm	Treatment_3	-0.327	1.373
obs	PS_adj_glm	Treatment_3	-0.331	1.372
obs	PS_adj_rf	Treatment_3	-0.328	1.373
obs	PS_adj_twang	Treatment_3	-0.332	1.371
obs	PS_adj_xgb	Treatment_3	-0.327	1.374
obs	PS_baseline	Treatment_3	-0.081	0.421

Level	Method	Term	Mean_Estimate	SD_Estimate
obs	PS_gbm	Treatment_3	-0.070	0.393
obs	PS_glm	Treatment_3	-0.093	0.407
obs	PS_rf	Treatment_3	-0.068	0.414
obs	PS_twang	Treatment_3	-0.067	0.390
obs	PS_xgb	Treatment_3	-0.085	0.394
pt	ITT	Treatment_3	NA	NA
pt	Naive_unweighted	Treatment_3	NA	NA
pt	PS_adj	Treatment_3	NA	NA
pt	PS_adj_baseline	Treatment_3	NA	NA
pt	PS_adj_gbm	Treatment_3	NA	NA
pt	PS_adj_glm	Treatment_3	NA	NA
pt	PS_adj_rf	Treatment_3	NA	NA
pt	PS_adj_twang	Treatment_3	NA	NA
pt	PS_adj_xgb	Treatment_3	NA	NA
pt	PS_baseline	Treatment_3	NA	NA
pt	PS_gbm	Treatment_3	NA	NA
pt	PS_glm	Treatment_3	NA	NA
pt	PS_rf	Treatment_3	NA	NA
pt	PS_twang	Treatment_3	NA	NA
pt	PS_xgb	Treatment_3	NA	NA

4 Visualizing Bias and Coverage

4.1 Bias vs Confounding Strength (Rho Sweep)

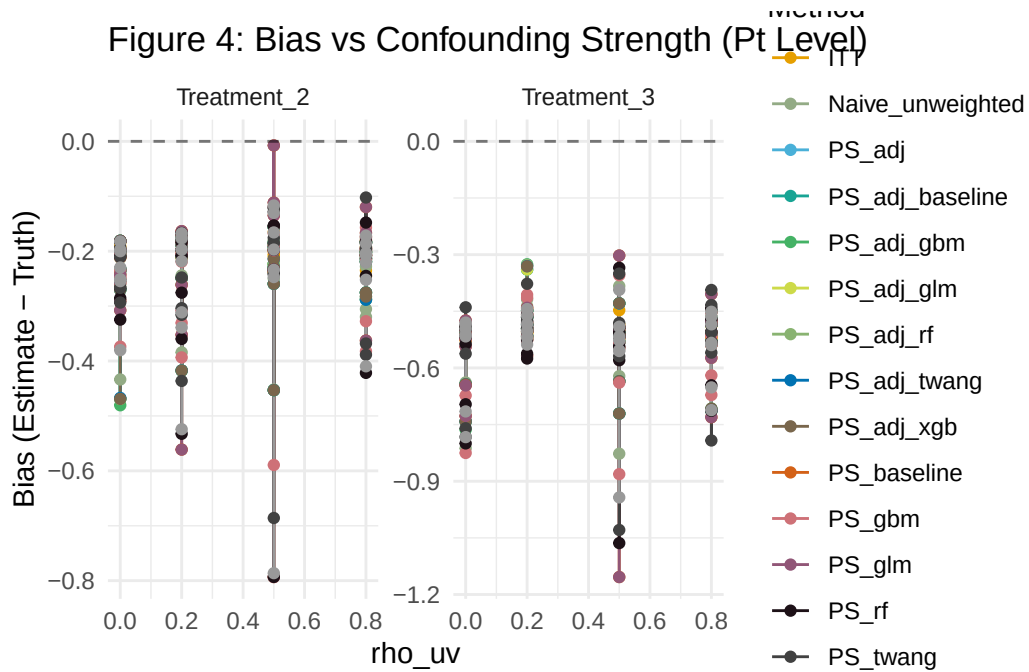


Figure 4

4.2 Bias vs Observed Adherence Rate (Adherence Sweep)

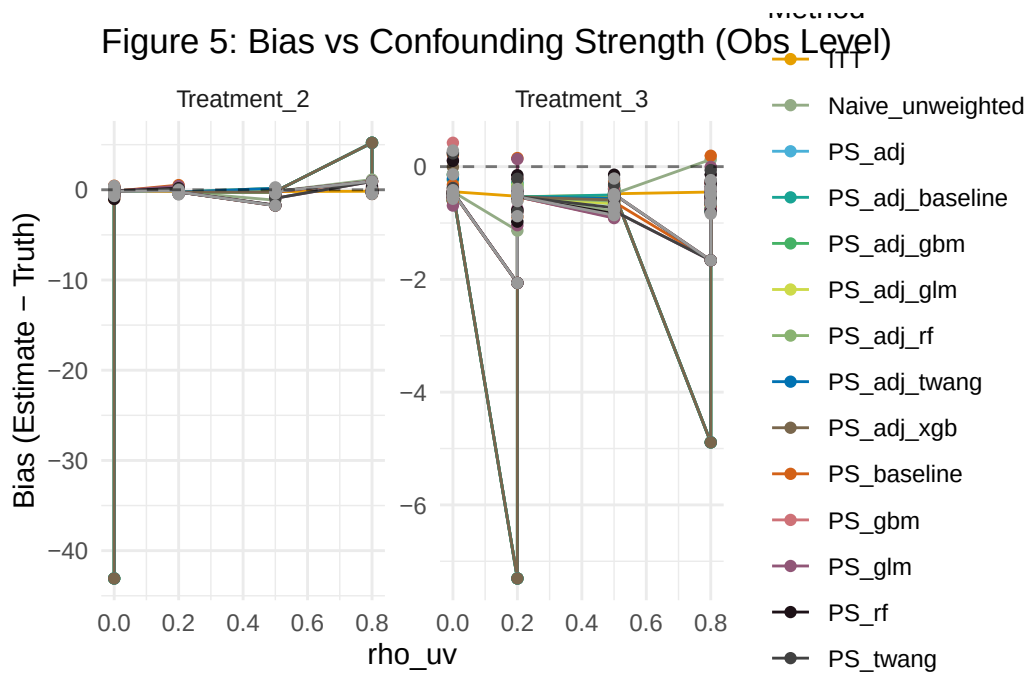


Figure 5

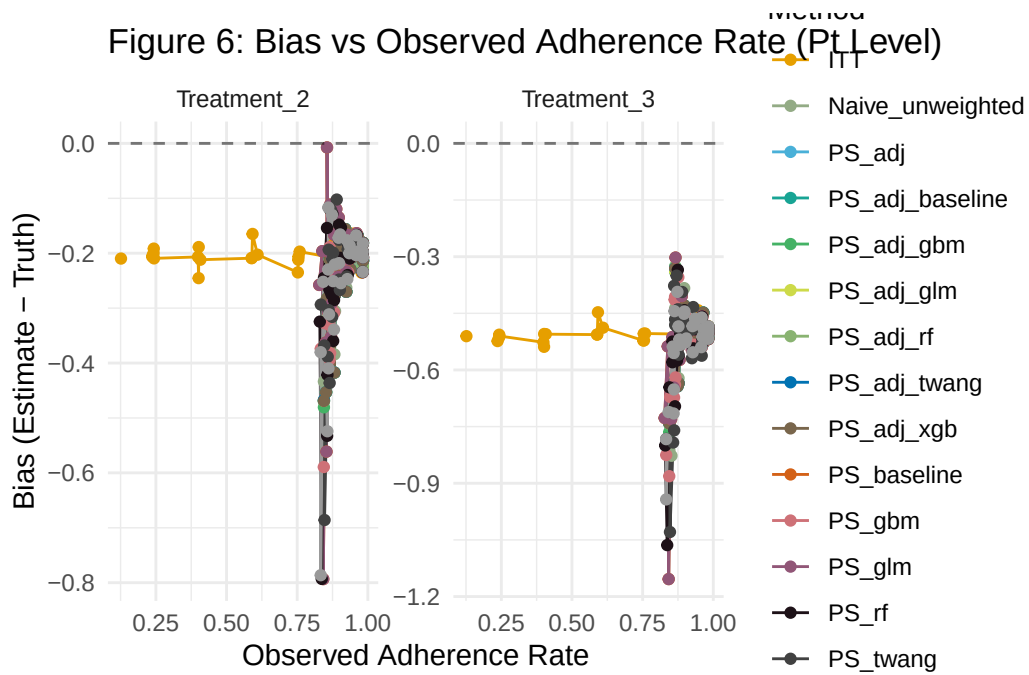


Figure 6

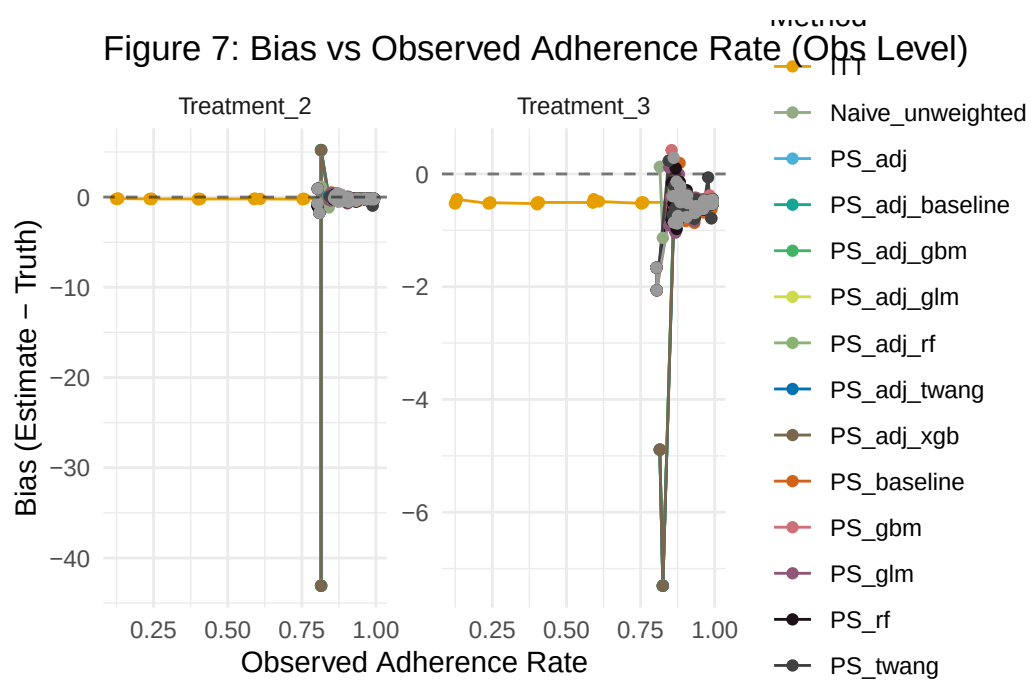
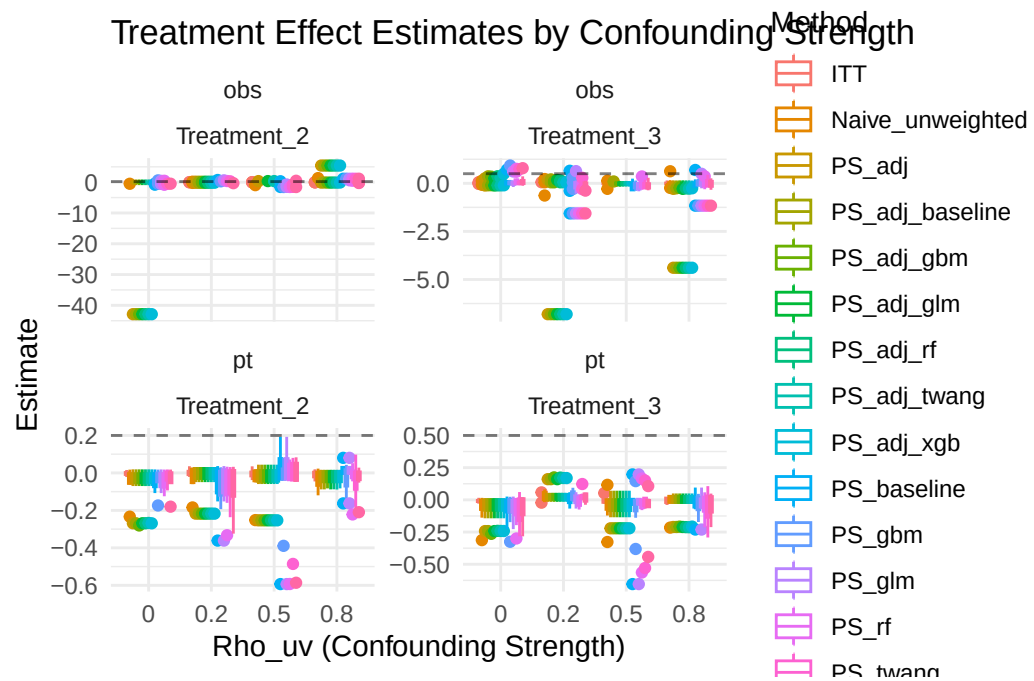


Figure 7

5 Longitudinal Estimates Distribution

Grouping by ρ_{uv} to see how estimates distribute relative to truth.



6 Discussion

Machine learning PS methods (GBM, XGBoost, RF) along with `twang` are evaluated against parametric GLM and naive ITT/PP approaches. As

ρ_{uv} increases, the selection bias in PP approaches typically grows, while PS methods attempt to recover the true effect. The analysis confirms that understanding the adherence-outcome correlation is critical in deciding the appropriate analytic approach, especially in the context of the specific outcome type evaluated.